

Pioneering Advancements in Biomedical Engineering through Brain-Computer Interface (BCI) Research

Background and Motivation

I am currently pursuing a Master's degree in Microelectronics Technology and Materials at The Hong Kong Polytechnic University, after a decade of professional experience in advanced technology fields. My career highlights include developing AI-based image processing algorithms, robotics control systems, and hardware-software integration solutions as an Algorithm Engineer and R&D Lead at prominent companies. My entrepreneurial journey, spanning four years, involved leading multidisciplinary teams and securing over ten invention patents, which enriched my ability to bridge technological innovation and practical applications.

The convergence of my interests in engineering, neuroscience, and biomedical applications has drawn me to Brain-Computer Interface (BCI) technology. BCI represents an interdisciplinary frontier with immense potential to transform healthcare, rehabilitation, and human-computer interaction. I am particularly motivated by the challenge of addressing technical barriers such as signal acquisition precision, data processing efficiency, and device usability. Pursuing a PhD at NUS will enable me to contribute meaningfully to this rapidly evolving field.

Why NUS for Biomedical Engineering?

NUS offers a unique ecosystem that blends world-class research facilities, interdisciplinary collaborations, and a strong focus on translational biomedical innovation. As a global leader in academic excellence, NUS provides access to cutting-edge research resources and opportunities to work alongside renowned faculty and researchers in the Biomedical Engineering Department.

Singapore's strategic emphasis on integrating advanced technology into healthcare aligns seamlessly with my research aspirations. The dynamic environment at NUS, characterized by its strong ties to industry and healthcare institutions, presents an ideal platform to translate research into impactful real-world applications.

Proposed Research Objectives

My research will focus on improving the precision and reliability of neural signal acquisition systems, developing AI-driven models for robust signal recognition, and exploring their applications in brain-controlled assistive devices. Specific objectives include:

1. **Enhancing Signal Acquisition:** Developing electrode materials and configurations that optimize signal strength, stability, and biocompatibility.
2. **AI-Driven Signal Processing:** Utilizing machine learning algorithms to refine neural signal recognition, addressing challenges in accuracy and noise reduction.

3. **Real-World Applications:** Integrating BCI technology into robotics, particularly for rehabilitation, by leveraging my expertise in robotics and AI to create adaptive, user-friendly systems.

Through interdisciplinary collaborations, I aim to bridge engineering, neuroscience, and clinical research to contribute novel solutions to the field of BCI.

Why I am Well-Suited for This Program

My background in microelectronics, AI, and robotics positions me uniquely to contribute to BCI research. My hands-on experience in designing and optimizing algorithms, hardware prototypes, and integrated systems equips me to tackle the multifaceted challenges in biomedical engineering. Additionally, my multilingual proficiency in English, Mandarin, and Cantonese will facilitate collaboration within the diverse academic and professional environment at NUS.

Long-Term Goals and Contributions

Upon completing my PhD, I aspire to become a leading researcher in biomedical engineering, specializing in BCI technology. My vision is to create innovative rehabilitation and healthcare solutions that enhance quality of life for individuals with disabilities. Furthermore, I aim to contribute to academia by fostering interdisciplinary collaboration and mentoring the next generation of researchers.

Conclusion

I am deeply inspired by the transformative potential of BCI technology and am eager to pursue this vision at NUS. With my technical expertise, professional experience, and passion for biomedical innovation, I am confident in my ability to contribute to the university's research ecosystem. Thank you for considering my application. I look forward to the opportunity to bring my skills and aspirations to NUS.